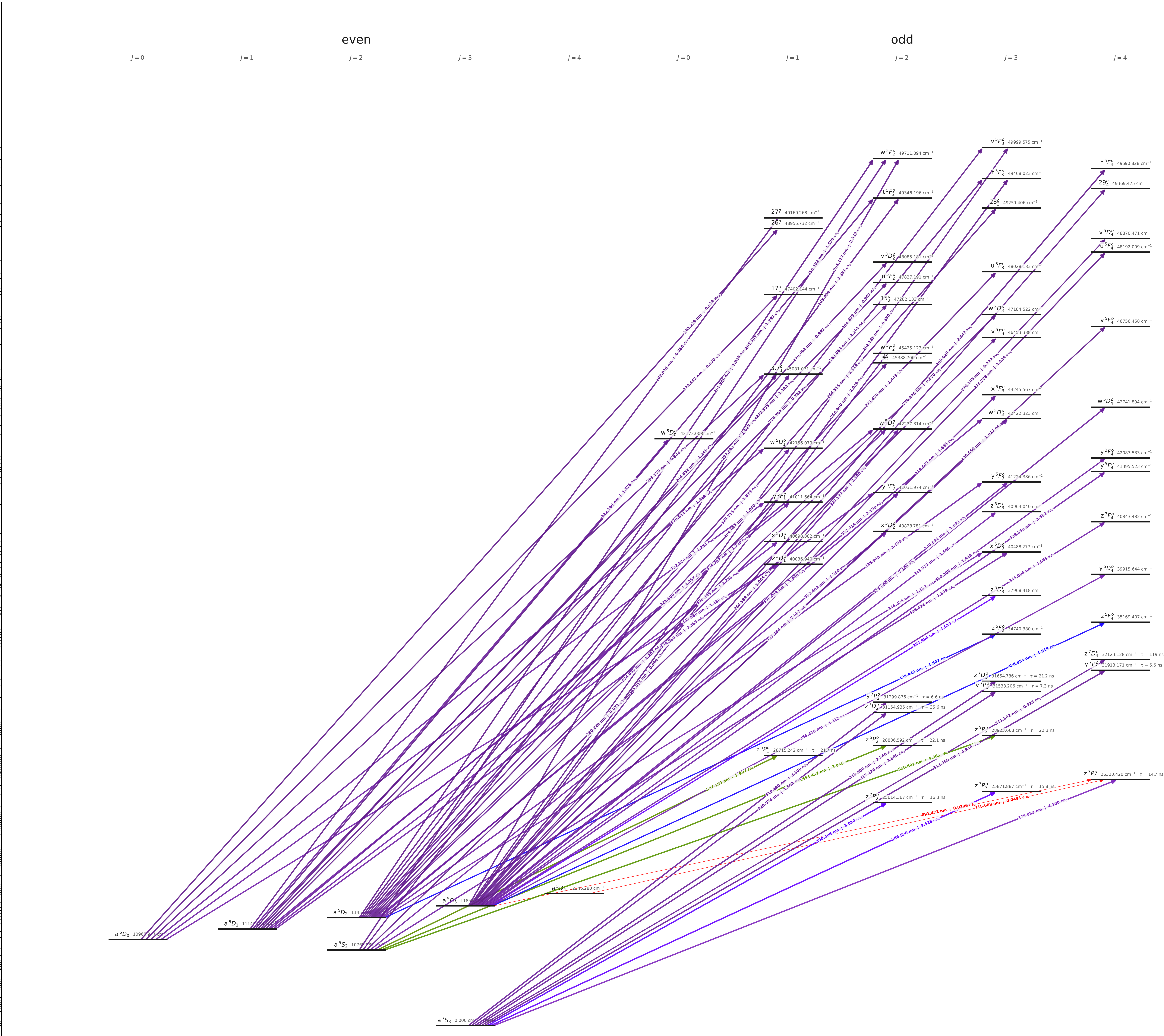


Mo (Mo I). Strongest classified transitions below 2 μm (NIST): 62 levels, 72 lines.

Mo⁺ ionisation limit 57204.30 cm⁻¹ = 7.09243 eV (174.812 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle V \| e r \| J' \rangle$ in a_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J J' \rangle $ (ea ₀)	A (s ⁻¹)	$f/2\pi$ up. (MHz)	branch (%)	Use
$a^3S_1 - z^{-1}P_2^o$	379.933	379.8255	789.0659(6)	4.100	6.900e+07	10.83	100	laser-cooling / MOT cycling transition
$a^3S_1 - z^1P_1^o$	386.5199	386.4104	775.6197	3.528	6.240e+07	10.07	98.6	member of the a753 - z7P resonance triplet
$a^3S_1 - z^1P_2^o$	390.4059	390.2953	767.8994	3.010	6.170e+07	9.764	100	member of the a753 - z7P resonance triplet
$a^3D_3 - z^1P_2^o$	691.4710	691.2803	433.5575	0.021	2.900e+02	10.83	0	second leak channel of the 380 nm cooling transition
$a^3D_4 - z^1P_2^o$	715.6084	715.4112	418.9367(28)	0.043	1.150e+03	10.83	0	MOT repump line

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^7S_3 - z^7P_0^4$	379.933	92-96	920.4(19)
$a^7S_3 - z^7P_0^4$	379.933	94-96	418.4(16)
$a^7S_3 - z^7P_0^4$	379.933	98-96	-289.4(16)
$a^7S_3 - z^7P_0^4$	379.933	100-96	-812.0(19)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$

⁹⁵Mo hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)				
a^7S_3	-208.582060(10)	0.03705(10)	F=1/2: +2085.84	F=3/2: +1772.96	F=5/2: +1251.49	F=7/2: +521.44	F=9/2: -417.18 F=11/2: -1564.36

Measured A, B constants; sources in the data file.